



Arduino Based RGB COLOUR CODE GENERATOR

A. SAMIYUDDHIN

Red, green and blue (RGB) are the basic colours for generating various other colours by mixing colours in a particular proportion. This technique is used in TVs, mobiles etc. Simi-

larly, Web designing using HTML or any other language requires the hex code of a particular colour in the design. Presented here is the circuit for learning the concept of generating colours and the respec-

tive hex codes of those colours.

Circuit and working

Circuit diagram of the RGB colour code generator is shown in Fig. 1. It is built around Arduino Uno board

(Board1), 16x2 LCD (LCD1), RGB LED (RGB1) and a few other components. Arduino board is the brain of the circuit that performs colour generation using pulse width modulation (PWM). Three potentiometers control intensity of the RGB LED. The 16x2 LCD wired in 4-bit mode is used to display the value of each colour in the first row and

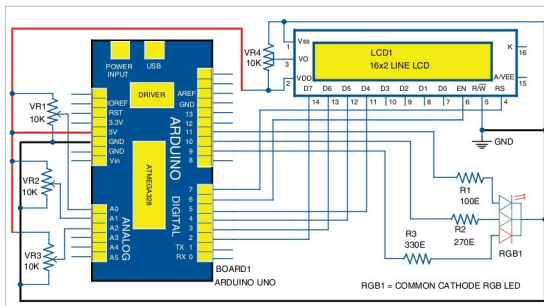


Fig. 1: Circuit of the RGB colour code generator



Fig. 2: Screenshot of the source code



Fig. 3: RGB colour display on the LCD

PARTS LIST

Semiconductors:

- | | |
|--------|--------------------------|
| Board1 | - Arduino Uno board |
| LCD1 | - 16×2 LCD |
| RGB1 | - Common-cathode RGB LED |

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- | | |
|---------|-----------------------------|
| R1 | - 100-ohm |
| R2 | - 270-ohm |
| R3 | - 330-ohm |
| VR1-VR3 | - 10-kilo-ohm potentiometer |
| VR4 | - 10-kilo-ohm preset |

Miscellaneous:

- CON1 - 5-pin connector
CON2 - 9-pin connector

the hex value in the second row,

Intensity of each colour is controlled by potentiometers VR1, VR2 and VR3. These are connected to analogue inputs A0, A1 and A2 of Arduino board. The program continuously scans the analogue input of the board. The analogue-to-digital converter (ADC) of Arduino converts the analogue value into a 10-bit digital value. Since resolution of the PWM module of the board is 8-bit, the 10-bit analogue value is converted into 8-bit value by the program. Screenshot of the source program is shown in Fig. 2.

Thus, the 8-bit value controls